

Campus Notification Engine

- Clean-room rebuild of a campus notification engine based on reverse-engineering observations of Novu.
- Goal: deliver high-volume campus notifications quickly without spam or duplicate deliveries.

Problem & Core Flow

- Event → workflow → preference check → delivery → activity/inbox tracking.
- Core requirements: in-app/email workflows, user preferences, burst digests, delivery tracking, and retries.

Killer Tests

- 10 events within 5 minutes → one digest.
- Email muted → in-app only.
- Failed send → retry without duplicate.

Our Improvements

- Gap fix: atomic workflow/control-value writes using a transaction/session.
- Differentiator: academic-calendar-aware digest scheduling.
- Exam periods can use a longer digest window than the default non-exam window.

Rebuild Direction

- Documented architecture, data model, API and acceptance criteria are in docs/.
- Implementation follows the clean-room specification.
- Original source code is not copied.